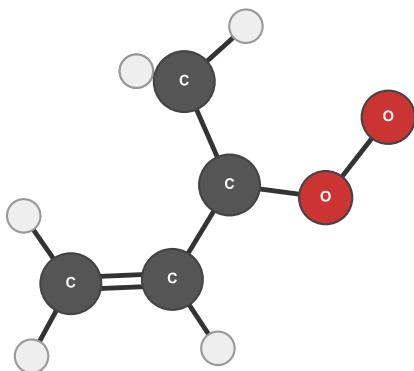


syn

trans
syn-trans



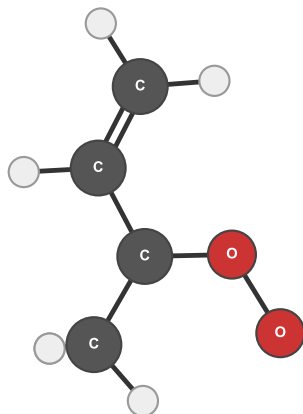
$\Delta E = +0.00 \text{ kcal mol}^{-1}$ (wB97X-D)

$\Delta E = 0.00 \text{ kcal mol}^{-1}$ (Barber CCSD(T))

CASSCF(2,2): $\Delta E_{S1}=23.4$, $\Delta E_{T1}=20.7 \text{ kcal mol}^{-1}$

wB97X-D opt

cis
syn-cis



$\Delta E = +1.81 \text{ kcal mol}^{-1}$ (wB97X-D)

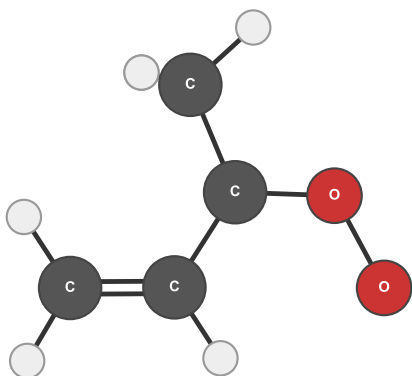
$\Delta E = 1.76 \text{ kcal mol}^{-1}$ (Barber CCSD(T))

CASSCF(2,2): $\Delta E_{S1}=21.6$, $\Delta E_{T1}=19.0 \text{ kcal mol}^{-1}$

wB97X-D opt

anti

anti-trans



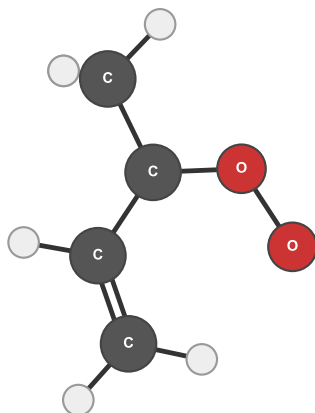
$\Delta E = +0.44 \text{ kcal mol}^{-1}$ (wB97X-D)

$\Delta E = 2.57 \text{ kcal mol}^{-1}$ (Barber CCSD(T))

CASSCF(4,4): $\Delta E_{S1}=27.3$, $\Delta E_{T1}=24.8 \text{ kcal mol}^{-1}$

wB97X-D opt

anti-cis



$\Delta E = -0.02 \text{ kcal mol}^{-1}$ (wB97X-D)

$\Delta E = 3.05 \text{ kcal mol}^{-1}$ (Barber CCSD(T))

CASSCF(4,4): $\Delta E_{S1}=29.0$, $\Delta E_{T1}=26.5 \text{ kcal mol}^{-1}$

wB97X-D opt

■ C ■ O ■ H